

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

MSE 4201
(Materials for Energy Conversion and Storage)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) "Energy and Materials have an inseparable links", Justify your answer. (09)
(b) How do photovoltaic cells convert sunlight to electricity? Explain with a suitable figure. (15)
(c) "Conductivity of silicon is temperature dependent"-Explain. (06)
- Q 2. (a) How do Dye-Sensitized solar cells differ from organic solar cells? (10)
(b) Explain the working principle with a necessary figure of an energy conversion fuel cell considered suitable for vehicles and mobile applications. (15)
(c) Why is a buffer layer so important in organic photovoltaics? (05)
- Q 3. (a) Which aspects need to meet during waste storage? (08)
(b) What is TRISO fuel? Briefly discuss about it. (12)
(c) Briefly discuss one of the environmental friendly fission waste disposal methods. (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) "Batteries and capacitors should be regarded as complementary technologies, each providing a unique solution to energy storage", Explain how. (05)
(b) Explain the working mechanism with necessary figure of an energy storage battery which is considered one of the most promising candidates for stationary electric energy storage. (12)
(c) Write a suitable energy storage battery which is a part of the molten salt battery family and describe the battery composition and construction. (13)
- Q 5. (a) Which metal has great promise as an electrode material for batteries among all the existing metals? Justify your answer. (06)
(b) Which batteries have comparatively greater energy storage efficiency and charge-discharge efficiency than other rechargeable batteries? Write down the fundamental principles of such batteries. (12)
(c) "Zinc/air batteries are metal/air batteries powered by oxidizing zinc with oxygen from the air". Explain with necessary figure. (12)

- Q 6. (a) "Pseudo-capacitor is a kind of super capacitor which is different from double layer capacitor"- do you agree or disagree with the statement? Justify your answer. (12)
- (b) If two capacitors $C_1 = 15 \text{ F}$ and $C_2 = 8 \text{ F}$ were connected in series and parallel to a 50 volt battery, determine the voltage across each capacitor for each type of connection. (08)
- (c) "A li-ion capacitor (LIC) is a hybrid type of capacitor classified as a type of super-capacitor"-do you agree or disagree with the statement? Justify your answer. (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

MSE 4211
(Professional Ethics)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What is Engineering Ethics? What are the scopes of it in professional practices. (10)
(b) State the code of Ethics and its limitations in Engineering perspective. (20)
- Q 2. (a) Please explain the different types of models of professional roles. (15)
(b) One of the moral issues in Engineering Ethics is Attitude toward Employees. Why poor attitude toward employees is set? (15)
- Q 3. (a) Differentiate between self-respect and self esteem. (05)
(b) Narrate the similarities and contracts of Engineering projects and standard experiments. (15)
(c) Explain the cost-benefit approach in utilitarian thinking. (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Distinguish between agency loyalty and identification loyalty. How does this classification differ from that of critical and uncritical loyalty? (12)
- (b) "It is impossible to endorse fully the collective bargaining of unions or to condemn. There exist always conflicting views between professionalism and unionism". Do you agree with this statement? Justify your answer. (12)
- (c) How does an engineer acquire authority by expertise? (06)
- Q 5. (a) Justify the statement that the engineer has the right for recognition for his excellence and achievements in his service. (10)
- (b) Examine the following illustration and see if it is the right situation for whistleblowing. (14)
- "A young engineer felt that the level of pollutants pouring into the nearby stream from his factory were dangerously high considering the fact that children were swimming down the stream. On informing this to his boss, he was told that his fears were baseless as there had been no complaint from any quarter in the past. What should the engineer do at this stage?".
- (c) Why should engineering information be kept confidential? (06)
- Q 6. (a) Who is likely to be harmed by fabricating research data? Does actual harm have to occur in order for fabrication to be ethically wrong? Explain briefly. (12)
- (b) How does patent differ from copyright. (06)
- (c) What is an engineer's role in protecting environment? Explain briefly. (12)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term (Regular) Examination, 2020
Department of Materials Science and Engineering

MSE 4231
(Biomaterials)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) 'Biomaterials is a field of Materials Science which is always evolving to achieve the highest level of biocompatibility'-Explain this statement for metallic biomaterials. (15)
- (b) What is meant by 'Stress shielding Effect'? Why is it an important factor in determining if a metal is suitable for biomedical implantation or not? (15)
- Q 2. (a) 'An implant that performs well in one region of the body may suffer an unacceptable amount of corrosion in another'-Explain. (15)
- (b) Comment on the usage of Titanium as a metallic biomedical implant? What are the advantages and disadvantages of using Titanium in this purpose? (15)
- Q 3. (a) Why are shape memory alloys used in orthodontics? (10)
- (b) How do biological tissues respond to biomedical implants? (10)
- (c) Among in vitro, ex vivo and in vivo techniques, which one do you think more appropriate to estimate the extent of biocompatibility of a material and why? (10)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Define smart hydrogel and give an example of thermal responsive hydrogel. Why hydrogels are important in Polymeric Biomaterials? (15)
- (b) What are the parameters that influence the degradation rate of biopolymers? (15)
- Q 5. (a) What are the properties that required for dental, ophthalmic, orthopedic, cardiovascular, and drug delivery applications of Polymeric Biomaterials? (10)
- (b) Among powder, coating, and bulk bioceramics, which form will you prefer for space filling and tissue regeneration applications? Justify your answer. (10)
- (c) Write down the formula for the ceramic commonly present in dental enamel? What are the advantages of synthetic HAp? (10)
- Q 6. (a) Describe the processing of inert bioceramic. What are the desired properties for implantable bioceramics? (20)
- (b) What are the biomedical applications of pyrolytic carbon? (10)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

EEE 4227
(Semiconductor Materials and Devices)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Why does energy band bend under an applied electric field? Why electrons have greater mobility than holes? (08)
- (b) Define Fermi Energy level. Show that, "Fermi Energy level of a semiconductor can be shifted using n-type and p-type doping but it remains exactly in the middle of the bandgap for an intrinsic semiconductor". (12)
- (c) For a particular semiconductor material $N_c = 1.5 \times 10^{18} \text{ cm}^{-3}$, $N_v = 1.3 \times 10^{19} \text{ cm}^{-3}$ and $E_g = \frac{\text{last two digit of your roll number}}{20}$ (10)
- i) Determine the intrinsic carrier concentration of the semiconductor at $T=300\text{K}$.
ii) Find the position of the intrinsic Fermi level with respect to the center of the bandgap.
- Q 2. (a) Drive the equation for depletion region width of p-n junction under no applied bias. (12)
- (b) Show the energy band diagrams for a p-n junction under the following condition: (08)
- i) Open circuit
ii) Forward bias
iii) Reverse bias
iv) Zener breakdown
- (c) A p-n junction diode has a concentration of 10^{17} acceptor atoms cm^{-3} on the p-side and a concentration of 10^{18} donor atoms cm^{-3} on the n-side. What will be the built in potential for the semiconductor materials Ge, Si and GaN? ($T=300\text{K}$) (10)
- Q 3. (a) Why biasing is necessary for a transistor? How can a BJT transistor be used as an amplifier? Explain with example. (10)
- (b) What is Q-point for a BJT transistor? Explain three ways to shift Q-point of a transistor and comment on the suitable Q-point for the operating condition. (10)
- (c) Describe the operating principles of a JFET. Differentiate between JFET and BJT. (10)

Section B

(Answer ANY TWO questions from this section in **answer script B**)

- Q 4. (a) Explain the operation of a n-channel E-MOSFET when (i) $V_{GS} = 0V$, (ii) $V_{GS} = +ve$ V, (10)
and (iii) at pinch off.
- (b) For the n-channel depletion type MOSFET of Fig. 4(b) determine (i) I_{DQ} , (ii) V_{GSQ} and (12)
(iii) V_{DS} .

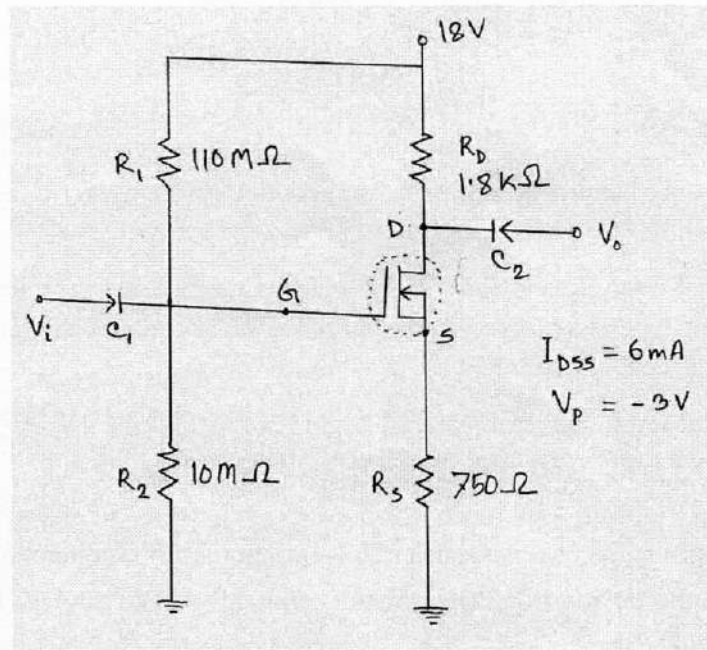


Fig. 4(b)

- (c) How can MOSFET be used as a switch? Explain briefly. (08)
- Q 5. (a) What is a Schottky barrier? Describe the operating principles of Schottky diode under (10)
following conditions with necessary energy diagram (i) Forward bias, and (ii) Reverse bias.
- (b) Write some differences between Schottky barrier diode and p-n junction diode. (08)
- (c) What is two dimensional electron gas (2DEG)? At thermal equilibrium, draw the ideal (12)
energy diagram of (i) nN heterojunction, and (ii) pP heterojunction.
- Q 6. (a) What happens when a photodiode is illuminated? Write down the factors that limit the (10)
speed of response of a photodetection.
- (b) What population inversion means? Draw the basic construction and explain the basic (12)
operating principles of a LASER.
- (c) How does DIAC work? Explain using characteristic curve. (08)

Khulna University of Engineering & Technology
B.Sc. Engineering 4th Year 2nd Term Examination, 2020
Department of Materials Science and Engineering

Hum 4227
(Project Management and Entrepreneurship)

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks

Section A

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Draw relationships of project risk and resource requirement at different stages of a project life-cycle. (15)
(b) How project management and project manager affect organizational value chain? Explain. (15)
- Q 2. (a) "Effective communication defines the success of a project"-Analyze. (10)
(b) Your company is considering two mutually exclusive projects whose costs and cash flows are shown below: (20)

Year	Project-A	Project-B
0	(10,000)	(11,000)
1	1,000	5,000
2	3,000	4,000
3	4,000	3,000
4	7,000	1,000

- (i) Calculate payback periods and IRR of the projects.
(ii) Which project should be accepted assuming cost of capital (K) is 10% ?
- Q 3. (a) Describe project selection criteria. How financial management is used to select a project. (15)
(b) Differentiate between PERT and CPM. What are the application of networking in project management? (15)

Section B

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) "Assuming the level of income remains unchanged, entrepreneurial initiatives reduce standards of living by introducing new needs"- Argue. (15)
(b) How entrepreneurs use financial statements to make the business models feasible? Explain (15)
- Q 5. (a) What is planning? Describe the need for planning. (15)
(b) State the role of entrepreneurship in moment of crisis. (15)

- Q 6. From the following Trial Balance of ABC Traders, prepare a statement of comprehensive income statement of owner's equity and a statement of financial position as on 31st December, 2020 (Considering the adjustments): (30)

ABC TRADERS TRIAL BALANCE As on 31 st December 2020		
Account Titles	Debit (Tk)	Credit (Tk)
Capital.....		5,00,000
Accounts Receivable	40,000	
Accounts Payable		20,000
Salaries expenses	30,000	
Prepaid Insurance	3000	
Rent expenses	60,000	
Utilities expenses	2000	
Gross Profit		1,80,000
Repair expense	1000	
Equipment	5,69,200	
Accumulated depreciation Equipment		5000
Bad debts expense	800	
Provision for doubtful debts		1000
	7,06,000	7,06,000

Additional information: (i) salaries unpaid Tk. 6000 for the period; (ii) Insurance expense expired per month Tk. 200. (iii) Rent expense per month Tk. 4000. (iv) Make allowance for doubtful debts on accounts receivable @ 3%.